EUN JI OH

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RESEARCH INTERESTS

I study how people perceive and evaluate music, with a focus on **the role of emotion and mood in musical reward**. My earlier work used behavioral and physiological experiments to compare how performers and listeners communicate emotion. Building on this, my current work combines computational modeling with behavioral data to examine how individual listeners construct the value of music from its acoustic, structural, and emotional features, and how these subjective representations drive reward. Next, I plan to extend this work with neural methods to understand mood regulation through musical reward. Ultimately, I aim to support well-being and mental health.

EDUCATION

Georgia Institute of Technology Aug 2025 – Present
Ph.D. in Music Technology (expected 2029) Atlanta, GA, USA

- Research focus: Computational modeling of musical reward, mood, and emotion
- Advisor: Dr. Claire Arthur (Computational and Cognitive Musicology Lab)

Korea Advanced Institute of Science and Technology (KAIST) Feb 2022 – Feb 2024
M.S. in Culture Technology Daejeon, Republic of Korea

- Thesis: *Shared empathic process in music and social contexts: Exploring empathic accuracy and physiological responses across modalities and valence*
- Advisor: Dr. Kyung Myun Lee (Music and Brain Lab)

Seoul National University Mar 2017 – Feb 2022
B.M. in Musicology (Minor in Psychology) Seoul, Republic of Korea

- Summa Cum Laude; GPA: 4.07/4.3

RESEARCH EXPERIENCE

Georgia Institute of Technology, School of Music Aug 2025 – Present
Graduate Research Assistant, Computational and Cognitive Musicology Lab Atlanta, GA, USA

- **Doctoral Project.** Developing computational models of music liking, a subjective evaluation shaped jointly by listeners' mood and the emotion expressed in music, using published datasets. Follow-up behavioral and functional magnetic resonance imaging (fMRI) studies planned.
- **AI Music Description.** Led the human-study component (online survey) of a collaborative project on how people use and evaluate large language model (LLM)-generated music descriptions in everyday life. (ISMIR 2026 Late-Breaking Demo)
- **Biofeedback Music.** Led two galvanic skin response (GSR)-based biofeedback systems for live performance: *The Singing Skin*, which maps audience GSR to rhythmic modulation (ICMC 2026), and *Bio Sing-Along*, which maps multiple audience members' GSR to timbre via a Differentiable Digital Signal Processing (DDSP) model (with J.W. Beck), presented at *PRISM Pt. 1* (May 2026). Funded by the Georgia Tech Center for Music Technology (GTCMT) Seed Grant.
- **Sharing Musical Experiences.** Investigated how and why people share the inner experiences music evokes (emotions, thoughts, memories), as distinct from sharing music itself. (SMPC & MUSICONNEX 2026)

KAIST, Center for Digital Humanities and Computational Social Sciences May 2024 – May 2025
Research Associate, Music and Brain Lab Daejeon, Republic of Korea

- **Music Pleasure.** Analyzed electroencephalography (EEG) interbrain synchrony (theta coherence) as a neural marker of shared musical emotion across valence.
- **X-Space.** Led a project developing a real-time EEG interbrain-synchrony system (phase locking value, PLV) for performer interaction, overseeing system concepts and development, and directed its live performance with Daejeon Arts Center.

KAIST, Graduate School of Culture Technology Feb 2022 – Feb 2024
Graduate Research Assistant, Music and Brain Lab Daejeon, Republic of Korea

- **Master's Thesis.** Examined real-time emotion inference across music and social contexts using piano improvisations (8 pianists, 36 listeners) presented in audio-only, video-only, and audiovisual conditions, with electrocardiography (ECG), continuous ratings, and linear mixed-effects models; found a music–social link for anger.
- **Emotion Communication & MIR.** Investigated which music information retrieval (MIR) audio features convey emotion from performers' expression to listeners' perception in piano improvisation, modeling time-varying audio features and continuous ratings with linear mixed-effects models and extending data from the master's thesis. (ISMIR 2024)
- **Musical Flow.** Reviewed the definition and structure of musical flow and assisted related EEG experiments.
- **Music and AI.** Prepared human-composed music stimuli and co-designed an online listening experiment evaluating AI-generated music in comparison with human-composed music.
- **Music Preference.** Assisted EEG experiments and prepared stimuli to investigate neural indicators of musical preference.

KAIST Jan 2021 – Feb 2022
Student Intern, Music and Brain Lab Daejeon, Republic of Korea

- **Music and Language.** Ran an online music-training program for 7-year-olds and assessed pre/post language and cognitive abilities.

Seoul National University Jan 2019 – Feb 2019
Student Intern, Music and Audio Research Group (MARG) Seoul, Republic of Korea

PUBLICATIONS

* Corresponding author

Conference Proceedings (peer-reviewed)

1. **Oh, E.J.***, Beck, J.W., and Smith, A. (2026). The Singing Skin: An audience-centered biofeedback system for musical interaction based on galvanic skin response. *Proceedings of the International Computer Music Conference (ICMC)*, Hamburg, Germany.
2. **Oh, E.J.**, Kim, H., and Lee, K.M.* (2024). Which audio features can predict the dynamic musical emotions of both composers and listeners? *Proceedings of the 25th International Society for Music Information Retrieval Conference (ISMIR)*, San Francisco, CA, USA, pp. 352–359.

Journal Articles (peer-reviewed)

1. **Oh, E.J.**, Park, J., Choi, Y., and Lee, K.M.* (2023). Understanding musical flow during music listening based on empirical observations. *Music Theory Forum*, 30(2), 101–140. (in Korean)
2. **Oh, E.J.**, and Lee, K.M.* (2020). Possibility of music training to promote Korean language development. *Journal of Music and Theory*, 35(1), 79–107. (in Korean)

Manuscripts in Preparation

1. **Oh, E.J.**, and Lee, K.M. An association between real-time emotion recognition ability in music and social-emotional contexts.

Conference Presentations

Oral Presentations

1. **Oh, E.J.**, and Lee, K.M.* (2024, July). Intermodal analysis of emotion inference: Examining shared processes in music and social contexts. *Conference of the Society for Music Perception and Cognition (SMPC)*, Banff, AB, Canada.

Poster Presentations

1. **Oh, E.J.***, Park, S., Kim, Y., and Beck, J.W. (2026, November). A user-centered view of AI music description. *International Society for Music Information Retrieval Conference (ISMIR), Late-Breaking Demo*, Abu Dhabi, United Arab Emirates.
2. **Oh, E.J.***, and Arthur, C. (2026, September). Understanding how people share musical experiences. *MUSICONNEX Conference*, Jyväskylä, Finland.
3. **Oh, E.J.***, and Arthur, C. (2026, July). Exploring the behavior of sharing musical experiences with others. *Conference of the Society for Music Perception and Cognition (SMPC)*, Evanston, IL, USA.
4. **Oh, E.J.**, and Lee, K.M.* (2024, March). Emotion inference during music and social situations through empathic accuracy. *67th Conference of the Korean Society for Music Perception and Cognition (KSMPC)*, Seoul, Republic of Korea. *(Best Poster Award)*
5. **Oh, E.J.**, and Lee, K.M.* (2023, August). Musical pleasure revealed by electroencephalography theta oscillations: A systematic review. *17th International Conference on Music Perception and Cognition (ICMPC)*, Tokyo, Japan.
6. **Oh, E.J.**, Kim, Y., and Lee, K.M.* (2023, August). Effect of online music class on linguistic, cognitive, and musical abilities in Korean preschool children: A pilot study. *17th International Conference on Music Perception and Cognition (ICMPC)*, Tokyo, Japan.
7. Kim, H., **Oh, E.J.**, Park, J., Chung, Y., Nam, J., and Lee, K.M.* (2023, August). A comparison of melody interpolation performed by human and artificial intelligence based on human similarity judgments. *17th International Conference on Music Perception and Cognition (ICMPC)*, Tokyo, Japan.

Grants & Fellowships

GTCMT Seed Grant ($2,000), Georgia Tech Center for Music Technology	Jan 2026
ISMIR Student Registration Grant ($650), 25th ISMIR Conference	Nov 2024
KAIST National Support & Master's Scholarships (KRW 41.6M)	2022 – 2023
SNU Work-Study & Merit Scholarships, Seoul National University	2019
Seongnam City Foundation Scholarship	2017

Honors & Awards

Arts Achievement Award, Georgia Tech Institute for Neuroscience, Neurotechnology, and Society	Apr 2026
Best Poster Award, 67th KSMPC Conference	Mar 2024
Summa Cum Laude, Seoul National University	Feb 2022

Music Technology Performances

SITE Festival: Graphic score performance; Technology Ensemble (MUSI 6004) — Oct 2026

- The Goat Farm, Atlanta, GA, USA

PRISM Pt. 1 — Sound through a New Lens: Interactive music and art event (GTCMT Seed Grant) — May 2026

- West Village, Georgia Institute of Technology, Atlanta, GA, USA

X-Space (KAIST × Daejeon Arts Center): Live performance with real-time EEG interbrain synchrony — Sep 2024

- Daejeon Arts Center, Daejeon, Republic of Korea

Teaching Experience

Teaching Assistant, KAIST, Daejeon, Republic of Korea

- HSS322: Understanding Music and the Brain, Prof. Kyung Myun Lee — Fall 2023
- GCT536: Cognitive Science of Music, Prof. Kyung Myun Lee — Spring 2023
- GCT565: Augmented Humans, Prof. Sang Ho Yoon — Fall 2022
- HSS186: Understanding Music and Music History, Prof. Kyung Myun Lee — Spring 2022

Service

Peer Review

- International Society for Music Information Retrieval Conference (ISMIR) — 2025, 2026
- Korean Society for Music Informatics (KSMI) — 2026

Academic & Community Service

- Student Secretary, Korean Society for Music Perception and Cognition (KSMPC) — Mar 2022 – Feb 2024
- Peer Counselor, Hug Program, KAIST — Mar 2023 – Feb 2024
- Program Assistant, Da Vinci Point Program (student arts engagement program), KAIST — Fall 2022

Skills

Programming	*R*: data analysis and statistical modeling
	Python: audio feature extraction (librosa, Essentia); windowed GSR feature computation with OSC output
	MATLAB: EEG analysis (EEGLAB); interbrain synchrony (theta coherence)
Statistical Analysis	Linear mixed-effects models, factor analysis (PCA, EFA, CFA)
Physiological Devices	EEG (BrainVision, Neuroscan CURRY 8, Emotiv), GSR/ECG (Shimmer)
Experiment Platforms	PsychoPy, Qualtrics, Prolific
Music Software	Finale, Logic Pro
Musicianship	Composition (Seoul Arts High School); piano (10+ years), violin, drums